

Data Collection, Exploratory Data Analysis, and Statistical Interpretation

Introduction

This stage of the project integrates data collection, exploratory data analysis (EDA), and statistical interpretation in order to understand the behavior of stock returns and identify the underlying drivers of portfolio performance.

The analysis is conducted using financial datasets collected from Yahoo Finance between 2018 and 2025. The dataset includes six major BIST-listed companies together with macroeconomic and market-wide variables such as the BIST100 index, USD/TRY exchange rate, and S&P500 index.

The primary purpose of this stage is to understand:

- Market dependency
- Volatility behavior
- Tail risk
- Correlation structures
- Portfolio sensitivity
- Statistical properties of financial returns

before applying machine learning models.

Dataset Structure

The dataset consists of the following Turkish stocks:

- THYAO
- GARAN
- AKBNK
- ASELS
- SISE
- EREGL

These companies were selected because they represent different sectors of the Turkish economy.

Additional datasets include:

- BIST100 Index (^XU100)
- USD/TRY Exchange Rate
- S&P500 Index (^GSPC)

The inclusion of macroeconomic and global variables allows the project to analyze both local and international financial influences on Turkish stock returns.

Data Collection and Preprocessing

Historical daily financial data was collected programmatically using:

- Yahoo Finance API
- yfinance Python library

The following variables were collected:

- Open Price
- High Price
- Low Price
- Close Price
- Adjusted Close Price
- Trading Volume

Adjusted closing prices were used for return calculations because they account for dividends and stock splits.

Several preprocessing steps were performed before analysis:

- Date alignment
- Missing value handling
- Infinite value cleaning
- Forward filling
- Return calculations
- Feature engineering

Daily returns were calculated instead of raw prices in order to:

- Normalize asset performance
- Improve comparability
- Reduce scale dependency between assets

The portfolio return was constructed as an equal-weighted average of the selected stocks.

Return Distribution Analysis

The distribution of returns was analyzed using:

- Histograms
- Boxplots
- Kernel density estimation

The results revealed that:

- Return distributions are non-normal
- Financial returns exhibit heavy tails
- Extreme observations occur frequently
- Volatility differs across sectors

ASELS generated the highest annualized return among the selected stocks, but also exhibited elevated volatility levels.

THYAO demonstrated strong returns together with the largest maximum drawdown, indicating high exposure to macroeconomic uncertainty and market shocks.

BIMAS displayed comparatively lower volatility and smaller drawdowns, suggesting that retail-sector firms behave more defensively during periods of economic stress.

The heavy-tailed nature of returns indicates that extreme gains and losses occur more frequently than predicted by normal distribution assumptions.

This finding is important because many traditional financial models assume normally distributed returns.

Cumulative Return Analysis

Cumulative return analysis was conducted to compare:

- Long-term portfolio performance
- Market benchmark behavior
- Sector dynamics

The equal-weighted portfolio closely followed the trajectory of the BIST100 index throughout the analyzed period.

This finding suggests that:

- Systematic market risk dominates firm-specific effects
- Turkish large-cap stocks move together under similar market conditions
- Local market behavior strongly influences portfolio returns

Although individual stocks occasionally deviated from market performance, the overall trend remained highly aligned with BIST100.

Correlation Analysis

A correlation heatmap was generated to examine relationships between:

- Individual stocks

- Portfolio returns
- BIST100 returns
- USD/TRY exchange rate
- S&P500 index

The analysis revealed:

- Strong positive correlations between most stocks and BIST100
- Significant market dependence among large-cap Turkish equities
- Lower correlation values for BIMAS compared to other assets

THYAO and GARAN exhibited especially strong market sensitivity, indicating that:

- Banking and aviation sectors are highly affected by macroeconomic conditions
- Systematic risk dominates sector behavior during volatile periods

The correlation structure also demonstrated that:

- Stocks frequently move together under similar market conditions
- Correlation alone does not imply predictability

Although strong contemporaneous relationships exist, these relationships may not generate reliable forecasts for future returns.

Volatility Analysis

Rolling volatility analysis was performed to examine:

- Time-varying risk
- Volatility persistence
- Market instability

Rolling volatility calculations revealed clear evidence of:

- Volatility clustering
- Persistent financial uncertainty
- Dynamic market behavior

Periods of high volatility were frequently followed by additional high-volatility periods, confirming one of the most important characteristics of financial time-series data.

THYAO and GARAN experienced particularly large volatility spikes during unstable periods, while BIMAS maintained relatively smoother volatility behavior.

This finding supports the idea that:

- Financial risk is not constant over time
- Market shocks create persistent uncertainty
- Investor behavior changes dynamically under stress

Rolling Beta Analysis

Rolling beta analysis was conducted to evaluate changes in market sensitivity over time.

Beta values were calculated relative to the BIST100 index in order to measure:

- Portfolio exposure to systematic market risk
- Dynamic market dependence

The analysis showed that:

- Beta values fluctuate significantly over time
- Market dependency increases during periods of instability
- Investors react more strongly to market-wide developments during uncertain conditions

During highly volatile periods:

- Correlations increased
- Market sensitivity strengthened
- Systematic risk became dominant

This confirms that Turkish equity markets become increasingly dependent on broader market conditions during periods of financial stress.

Drawdown and Downside Risk Analysis

Maximum drawdown analysis was performed to evaluate downside risk exposure.

The results demonstrated that:

- THYAO experienced the largest drawdown
- BIMAS exhibited relatively lower downside risk
- High-return assets often involve substantial short-term losses

This finding emphasizes the importance of:

- Risk management
- Portfolio stability
- Investor psychology

Large drawdowns are particularly important because temporary losses may significantly influence long-term investment behavior and decision-making.

Volume and Volatility Relationship

The relationship between trading volume and volatility was examined using scatter plots comparing:

- Average trading volume
- Absolute return magnitude

The results revealed a positive relationship between:

- Trading activity
- Volatility

Higher trading volume was generally associated with larger price movements.

This finding is consistent with financial theory, where:

- High trading activity reflects increased uncertainty
- Information flow intensifies during volatile periods
- Investor reactions amplify market fluctuations

Hypothesis Testing

Several statistical tests were conducted to evaluate whether observed relationships were statistically significant.

One-Sample t-Test

The one-sample t-test examined whether mean stock returns differed significantly from zero.

Results showed:

- Most p-values exceeded the 0.05 significance level
- The null hypothesis could not be rejected for most assets

This finding supports the weak form of market efficiency.

Paired t-Test

A paired t-test compared portfolio returns with BIST100 returns.

Results indicated that:

- The portfolio did not significantly outperform the market
- Market-wide dynamics dominate stock behavior

This finding further confirms the strong dependency between Turkish large-cap stocks and the broader market.

Pearson Correlation Test

Pearson correlation tests examined the relationship between:

- Trading volume
- Volatility

The results were statistically significant, allowing rejection of the null hypothesis.

This confirms that:

- Increased trading activity is associated with higher volatility
- Market uncertainty intensifies during periods of elevated volume

Regression Analysis

Multi-factor regression models were implemented to quantify relationships between:

- Portfolio returns
- BIST100 returns
- USD/TRY returns
- S&P500 returns

The regression results showed that:

- BIST100 possesses the strongest explanatory power
- Local market conditions dominate portfolio behavior
- USD/TRY and S&P500 exert weaker but still meaningful effects

The findings suggest that:

- Systematic market exposure is stronger than firm-specific effects
- Turkish stocks remain highly dependent on domestic market conditions

The R-squared values also indicated that:

- A significant portion of return variability can be explained by market-wide factors
- Some return variation remains unexplained due to the complex nature of financial markets

Implications for Machine Learning Models

The EDA findings revealed several characteristics of financial time series:

- Volatility clustering
- Heavy-tailed distributions
- Dynamic market dependency

- Time-varying risk

These findings suggest that:

- Financial data exhibits non-linear behavior
- Traditional linear models may struggle to capture complex relationships
- Machine learning models may better adapt to changing market conditions

The presence of non-linear dynamics justified the transition to:

- Logistic Regression
- Random Forest
- Gradient Boosting

in the next stage of the project.

Integrated Interpretation

The exploratory data analysis, hypothesis testing, and regression analysis together form a coherent interpretation of Turkish stock market behavior.

The results consistently demonstrate that:

- Turkish large-cap stocks are strongly influenced by BIST100
- Financial returns exhibit heavy tails and volatility clustering
- Market-wide dynamics dominate stock behavior
- Macroeconomic variables influence returns but remain secondary to local market conditions

The findings also highlight that:

- Strong correlation does not necessarily imply predictability
- Financial markets remain highly uncertain and dynamic

Conclusion of EDA Stage

The exploratory analysis demonstrates that Turkish stock returns are highly dependent on:

- Market-wide dynamics
- Volatility conditions
- Systematic risk exposure

Although macroeconomic and global variables contribute to return behavior, their effects remain secondary compared to the influence of the local market.

Overall, the results indicate that:

- Financial markets are non-linear and dynamic
- Risk behavior changes over time
- Predictive modeling requires more advanced techniques than traditional linear approaches

These findings provide the analytical foundation for the machine learning stage of the project.